

E.3 / PAPER 2 / QUESTION 5

30 April 2025 / TZ1 (file label) / HL Paper 2 / original Q3 / 2225-6709

5. (a) Rutherford modelled the scattering of alpha particles by a nucleus of gold. However, when the initial alpha particle energy is greater than a certain value E_0 , the fraction of alpha particles scattered at angles close to 180° does not follow Rutherford's model.

Outline

- (i) why this deviation from the model occurs, and [1]

.....
.....

- (ii) how E_0 can be used to estimate the nuclear radius of gold. [2]

.....
.....
.....
.....

Gold-198 (Au-198) is a radioactive nuclide of gold. Au-198 decays into stable mercury-198 (Hg-198) with an emission of a beta particle followed by one or two gamma photons.

The half-life of Au-198 is 2.69 days.

- (b) (i) Calculate, in s^{-1} , the decay constant of Au-198. [1]

.....
.....

- (ii) A sample contains 5.0 mg of pure Au-198. Determine the mass of Hg-198 present in the sample after one week. [2]

.....
.....
.....
.....

(This question continues on the following page)

E.3 / PAPER 2 / QUESTION 5

30 April 2025 / TZ1 (file label) / HL Paper 2 / original Q3 / 2225-6709

(Question 5 continued)

Gamma photons of the following frequencies are emitted in the decay of Au-198:

Frequency / 10^{20} Hz
1.00
1.63
2.63

- (c) Explain how this observation provides evidence that nuclear energy levels are discrete. [2]

.....
